

8051 Microcontroller Architecture

1 8051 Overview

8051 Microcontroller

The **Intel 8051** is an 8-bit microcontroller introduced in 1980. It has become one of the most popular and widely studied microcontroller architectures due to its simplicity and versatility. The 8051 uses a **Harvard architecture** with separate program and data memory spaces.

1.1 8051 Architecture Components Summary

Component	Size/Count	Purpose
CPU	8-bit	Instruction execution and control
Internal ROM	4 KB	Program storage
Internal RAM	128 bytes	Data storage
I/O Ports	4 ports (32 pins)	External interfacing
Timers	2 (16-bit each)	Time delays and event counting
Serial Port	1 (Full-duplex)	UART communication
Interrupts	5 sources	Event-driven processing

2 Special Function Registers (SFRs)

SFR Register Bank

The **Special Function Register (SFR) bank** is a set of registers with addresses in the range **80H to FFH**. These registers control the operation of various hardware components including I/O ports, timers, serial port, and interrupts.

2.1 Accumulator (ACC or A)

Accumulator Register

The **Accumulator (A)** is the most important register in the 8051. It serves as the primary **operand register** for most arithmetic and logic operations.

Key Characteristics:

- **Size:** 8 bits wide.
- **Location:** Address E0H in the SFR bank.
- **Role:** Acts as an **implicit or specified operand** in instructions.
- **Usage:** Stores operands and results of arithmetic, logic, and data transfer operations.

Accumulator Usage Examples

```
MOV A, #55H      ; Load immediate value 55H into A
ADD A, R0        ; Add R0 to A, result stored in A
SUBB A, #10H     ; Subtract 10H from A with borrow
ANL A, #0FH      ; AND A with 0FH, result in A
```

Most operations use the accumulator either explicitly or implicitly.

2.2 B Register

B Register

The **B register** is a special 8-bit register primarily used for **multiply** and **divide** instructions. It can also serve as a **general-purpose scratch pad register** for other operations.

Key Characteristics:

- **Size:** 8 bits wide.
- **Location:** Address F0H in the SFR bank.
- **Primary Use:** Stores one of the operands for MUL and DIV instructions.
- **Secondary Use:** Used as a **scratch pad** (temporary register) in other instructions.

Scratch Pad Register

A **scratch pad register** is a **general-purpose temporary register** used to:

- Hold intermediate values during calculations.
- Store temporary data during complex operations.
- Act as a buffer when registers are limited.

B Register in Multiplication and Division

Multiplication (MUL AB):

```
MOV A, #25      ; Load 25 into A
MOV B, #10      ; Load 10 into B
MUL AB          ; A × B = 250 (Low byte in A, High byte in B)
```

Division (DIV AB):

```
MOV A, #100     ; Load 100 (dividend) into A
MOV B, #7       ; Load 7 (divisor) into B
DIV AB          ; Quotient (14) in A, Remainder (2) in B
```

2.3 Program Status Word (PSW)

Program Status Word

The **Program Status Word (PSW)** is an 8-bit register that contains a **set of flags** with **status information** about the current state of the CPU after executing instructions. It is essential for conditional branching and program control.

Location: Address D0H in the SFR bank.

2.3.1 PSW Bit Structure

Bit	Position	Name	Description
7	PSW.7	CY (Carry)	Set if there is a carry out of bit 7 during arithmetic operations.
6	PSW.6	AC (Auxiliary Carry)	Set if there is a carry from bit 3 to bit 4 (BCD arithmetic).
5	PSW.5	F0 (Flag 0)	General-purpose user flag for software control.
4	PSW.4	RS1	Register bank select bit 1.
3	PSW.3	RS0	Register bank select bit 0.
2	PSW.2	OV (Overflow)	Set if there is an overflow in signed arithmetic.
1	PSW.1	–	Reserved (user-defined flag).
0	PSW.0	P (Parity)	Set if the accumulator has an odd number of 1s.

2.3.2 Register Bank Selection

Register Banks

The 8051 has 4 register banks (Bank 0–3), each containing 8 registers (R0–R7). The **RS1** and **RS0** bits select the active bank:

RS1	RS0	Bank Selected	Address Range
0	0	Register Bank 0	00H – 07H
0	1	Register Bank 1	08H – 0FH
1	0	Register Bank 2	10H – 17H
1	1	Register Bank 3	18H – 1FH

3 Pointer and Address Registers

3.1 Stack Pointer (SP)

Stack Pointer

The **Stack Pointer (SP)** is an **8-bit wide register** that contains the **8-bit stack top address**. It points to the top of the stack in internal RAM.

Key Characteristics:

- **Size:** 8 bits wide.
- **Location:** Address 81H in the SFR bank.
- **Initial Value:** Initialized to 07H after reset.
- **Stack Location:** Stack can be defined **anywhere in the on-chip 128-byte RAM**.

Stack Operations

PUSH or CALL Instructions:

1. SP is **incremented by one** first.
2. Data is then **stored onto the stack** at the new SP address.

POP or RET Instructions:

1. Data is **read from the stack** at current SP address.
2. SP is then **decremented by one**.

Stack Pointer Initialization

After reset, SP = 07H means the stack starts at address 08H (first push). Since Register Bank 1 occupies 08H–0FH, it's common practice to initialize SP to a higher value (e.g., 2FH or 7FH) to avoid overwriting register banks.

3.2 Data Pointer (DPTR)

Data Pointer

The **Data Pointer (DPTR)** is a **16-bit register** used to hold **16-bit external data RAM addresses**. It consists of two 8-bit registers: **DPH** (higher byte) and **DPL** (lower byte).

Key Characteristics:

- **Size:** 16 bits (accessed as one 16-bit or two 8-bit registers).
- **Components:**
 - **DPH** (Data Pointer High) – Address 83H
 - **DPL** (Data Pointer Low) – Address 82H
- **Purpose:** Contains 16-bit addresses for accessing external data memory.
- **Addressing Range:** Can address up to **64 KB** of external memory (0000H – FFFFH).

DPTR Usage

```
MOV DPTR, #2000H    ; Load 16-bit address 2000H into DPTR
MOVX A, @DPTR        ; Read external RAM at address 2000H into A
MOVX @DPTR, A        ; Write A to external RAM at address 2000H
```

4 I/O Ports

4.1 Port 0 to Port 3 Latches and Drivers

I/O Ports

The 8051 has **4 on-chip I/O ports** (P0, P1, P2, P3), each 8 bits wide, providing a total of **32 I/O pins**. Each port has associated **latches and drivers** with addresses in the SFR bank.

Port	SFR Address	Alternate Functions
P0	80H	Address/Data bus (AD0–AD7) for external memory
P1	90H	General-purpose I/O
P2	A0H	Higher address bus (A8–A15) for external memory
P3	B0H	Special functions (RXD, TXD, INT0, INT1, T0, T1, WR, RD)

Port Communication

Port communication occurs through the **allotted SFR addresses**. Writing to a port address updates the port latch. Reading from a port address reads the pin or latch value depending on the instruction used.

5 Serial Communication

5.1 Serial Data Buffer (SBUF)

Serial Data Buffer

The **Serial Data Buffer (SBUF)** internally contains **two independent registers** that share the same address (99H):

- **Transmit Buffer:** A **parallel-in-serial-out** register for sending data.
- **Receive Buffer:** A **serial-in-parallel-out** register for receiving data.

Operation:

- **Writing to SBUF:** When a byte is written to SBUF, it **initiates serial transmission**. The data is loaded into the transmit buffer and shifted out bit by bit.
- **Reading from SBUF:** When SBUF is read, it **reads the received serial data** from the receive buffer.

SBUF Usage

```
MOV SBUF, A      ; Write A to SBUF, starts transmission

; After reception complete (RI flag set):
MOV A, SBUF      ; Read received byte from SBUF into A
```

6 Timer Registers

Timer Registers

The 8051 has **two 16-bit timer registers** (Timer 0 and Timer 1). Each timer is accessed as two separate 8-bit registers: **lower byte (TL)** and **upper byte (TH)**.

Register	SFR Address	Description
TL0	8AH	Timer 0 Lower Byte
TH0	8CH	Timer 0 Higher Byte
TL1	8BH	Timer 1 Lower Byte
TH1	8DH	Timer 1 Higher Byte

7 Control Registers

Control Registers

Control registers determine how certain hardware components work. They contain control and status information for **interrupts, timers/counters, and serial port**.

Register	Address	Function
IP	B8H	Interrupt Priority register – sets priority levels
IE	A8H	Interrupt Enable register – enables/disables interrupts
TMOD	89H	Timer Mode register – configures timer operating modes
TCON	88H	Timer Control register – timer run control and interrupt flags
SCON	98H	Serial port mode Control register – serial port configuration
PCON	87H	Power Control register – power-down and idle modes

8 Internal Architecture Units

8.1 Timing and Control Unit

Timing and Control Unit

The **Timing and Control Unit** derives all necessary **timing and control signals** required for:

- **Internal operation** of the circuit (fetching, decoding, executing).
- **Controlling the external system bus** for memory access.
- **Execution of instructions** in proper sequence.

8.2 Oscillator

Oscillator

The **Oscillator** generates the **basic timing clock signal** for the entire microcontroller. It uses a **crystal oscillator** connected to XTAL1 and XTAL2 pins (typically 11.0592 MHz or 12 MHz).

Machine Cycle

One **machine cycle** in 8051 = 12 oscillator periods. For a 12 MHz crystal:

- Machine cycle time = $\frac{12}{12 \text{ MHz}} = 1\mu s$

8.3 Instruction Register

Instruction Register

The **Instruction Register** holds the current instruction being executed. It **decodes the opcode** of the instruction and provides information to the **timing and control unit** for proper execution.

8.4 EPROM and Program Address Register

Program Memory Unit

The **EPROM and Program Address Register** provide:

- **On-chip EPROM (4 KB):** Stores the program code.
- **Program Address Register:** Mechanism to address the program memory (holds the address of the next instruction to fetch).

8.5 RAM and RAM Address Register

Data Memory Unit

The **RAM and RAM Address Register** provide:

- **Internal 128 bytes of RAM:** For data storage.
- **RAM Address Register:** Mechanism to address the internal RAM.

8.6 Arithmetic Logic Unit (ALU)

ALU

The **Arithmetic Logic Unit (ALU)** performs **8-bit arithmetic and logical operations**. Operands are held by temporary registers **TMP1** and **TMP2** before the operation.

ALU Operations:

- **Arithmetic:** Addition, Subtraction, Multiplication, Division, Increment, Decrement
- **Logical:** AND, OR, XOR, NOT, Rotate, Swap
- **Comparison:** Compare operations that affect flags

8.7 Interrupt, Serial Port, and Timer Units

Peripheral Control Units

The **Interrupt, Serial Port, and Timer units** are hardware blocks that **control and perform specific functions**. They operate **under the control of the timing and control unit** and can generate interrupts when specific events occur.

9 Summary: 8051 Architecture

Quick Reference Summary

Key Registers:

- **ACC (E0H):** Primary operand register for arithmetic/logic.
- **B (F0H):** MUL/DIV operand, scratch pad register.
- **PSW (D0H):** Status flags (CY, AC, OV, P) and bank selection (RS1, RS0).
- **SP (81H):** 8-bit stack pointer, initialized to 07H.
- **DPTR:** 16-bit pointer (DPH + DPL) for external memory.

I/O and Communication:

- **P0–P3:** Four 8-bit I/O ports with SFR addresses.
- **SBUF:** Dual transmit/receive buffer for serial communication.

Control Registers:

- **IE, IP:** Interrupt enable and priority.
- **TMOD, TCON:** Timer mode and control.
- **SCON, PCON:** Serial and power control.

Internal Units:

- **ALU:** 8-bit arithmetic/logic operations.
- **Timing & Control:** Generates all control signals.
- **Oscillator:** Basic clock generation.

The 8051 architecture provides a robust foundation for understanding embedded systems and microcontroller programming.